KWA-NXUSA HIGH SCHOOL

Technology Plan 2026 - 2027

Thembisile Buthelezi
702836960

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1. Overview of KwaNxusa High School

KwaNxusa High School is a no-fee public school located in the King Cetshwayo District, in Mthonjaneni Municipality (Melmoth), KwaZulu-Natal in South Africa. The school serves learners from Grade 8 to 12 and is classified as a quintile 3 rural school. Most learners come from communities with limited access to technology at home, and the school itself has very few digital resources.

There are no computer labs, no internet for everyday learning, and teachers do not use technology as part of their daily instruction. Science teachers rely almost completely on chalkboards, textbooks, and explanations without practical demonstrations. Because of limited lab equipment, it is difficult to show experiments or let learners experience hands-on science. This affects how learners understand topics in Life Sciences, Chemistry, and Physics, especially abstract or practical-heavy concepts.

This technology plan aims to introduce basic, accessible, and realistic tools that teachers at KwaNxusa High School can use to improve science teaching and learning. The main tool will be Google Classroom. This will make it easier for teachers to share notes, assignments, videos, and resources, even if learners access them on their phones. To support practical learning, the plan also includes PhET simulations, virtual labs, and virtual museum tours, which help learners see and interact with scientific concepts even when physical labs are not available.

This plan is meant to support teachers, improve learner engagement, and show how technology can be used in a rural school with limited resources. The plan is realistic and focuses only on tools that can actually be used in this setting.

1.1 Focus of the Technology Plan

The focus of this plan is on integrating technology into the teaching of science subjects (Life Sciences, Chemistry, and Physics) for Grade 10–12. Google Classroom will serve as the main platform for communication and sharing learning materials. PhET simulations, virtual labs, and virtual museum tours will be used to help learners understand scientific content in an interactive and visual way.

1.2 Why This Area Was Chosen

Rural schools like KwaNxusa High School are not exposed to technology in teaching. Most Science learning is theoretical, and learners hardly ever see demonstrations or experiments. Teachers have no digital tools to support lessons, and there is no platform for collaboration or sharing resources.

Introducing technology in science will help:

- Make lessons more engaging
- Improve understanding of difficult or abstract ideas
- Expose learners to digital tools they have never used before
- Give teachers better ways to present content
- Bring practical experiences into the classroom through simulations and virtual labs

2. Specific Area Within the School the Technology Plan Is Focused On

This technology plan focuses on the Science Department at the school, specifically the teaching of Life Sciences, Chemistry, and Physics for Grade 10–12 learners. These subjects were chosen because they rely heavily on practical demonstrations, laboratory activities, and visual explanations, which the school currently cannot provide due to lack of resources.

The plan targets:

- Science teachers who need tools to teach abstract and practical concepts more effectively.
- Grade 10–12 learners who need access to better learning materials and interactive content.
- Classroom teaching practices, helping teachers move from traditional textbook-only teaching to using simple digital tools that support CAPS-aligned content.
- Assessment and communication processes, which will be managed through Google Classroom to help teachers track assignments and share feedback more consistently.

Through this plan, the school can start with one area that urgently needs support and then expand to other subjects in the future once teachers are comfortable using technology.

3. Goals / Desired Outcomes

The main goals set to be achieved through this plan are discussed below. These goals focus on helping teachers integrate technology into their lessons and supporting learners with resources that improve understanding.

3.1 Improve access to digital learning materials

By the end of one academic year, all Grade 10–12 Science teachers will be using Google Classroom to share notes, assignments, announcements, and learning resources with their learners.

Outcome:

- Learners will have one digital space where they can access all Science materials.
- Teachers will reduce reliance on photocopies and notebooks.

3.2 Enhance understanding of scientific concepts through interactive tools

By June of the implementation year, teachers will integrate PhET simulations and virtual labs into at least two lessons per topic in Life Sciences, Chemistry, and Physics.

Outcome:

- Learners will engage with content visually and interactively.
- Teachers will be better able to demonstrate abstract or practical ideas without physical lab equipment.

3.3 Develop teachers' confidence in using technology

By the end of Term 2, all Science teachers will receive basic training on:

- Setting up and managing a Google Classroom
- Uploading materials and creating assignments
- Using PhET simulations and virtual labs
- Safe and responsible use of digital tools

Outcome:

- Teachers will feel confident and supported when using technology.

- Consistency across the Science Department will improve.

4.4 Improve learner engagement and participation

Throughout the year, teachers will use digital tools (Google Classroom, simulations, virtual tours) to increase learner participation during lessons and to provide opportunities for self-paced learning.

Outcome:

- Learners will show increased interest in science lessons.
- Teachers will be able to track participation through classroom observations and submitted assignments.

3.5 Support equitable access to science learning

By the end of the first year, the school will identify learners who have limited access to devices or data and will implement low-data or offline alternatives, such as:

- Downloaded simulations used offline
- Notes saved in PDF for easy phone access
- Printed backup materials for those with no device

Outcome:

- No learner will be excluded from the technology plan.
- The project will support fairness and inclusivity.

4. Associated Professional Standards

This technology plan is guided by a combination of international and local professional standards, including the International Society for Technology in Education (ISTE) Standards for Educators and Students, the South African Council for Educators (SACE) Professional Teaching Standards, and the Association for Educational Communications and Technology (AECT) Standards. (ISTE, 2017; SACE, 2018; AECT, 2023).

The ISTE Standards provide a framework for integrating technology in ways that enhance teaching and learning. For educators, the standards emphasise continuous professional growth, designing meaningful and inclusive learning experiences, modelling responsible digital citizenship, and using technology to monitor and assess student progress. For learners, ISTE promotes empowerment, knowledge construction, and responsible participation in digital spaces. These principles support the integration of technology to make science lessons more interactive, accessible, and learner-centred (ISTE, 2017).

The SACE Professional Teaching Standards offer a local perspective, outlining the professional, ethical, and practical expectations for South African teachers. These standards emphasise appropriate teaching strategies, the creation of inclusive learning environments, and ongoing professional development. In the context of rural schools, these standards reinforce the importance of adapting technology use to limited resources, supporting equity, and promoting effective Science teaching across Life Sciences, Chemistry, and Physics (SACE, 2018).

The AECT Standards provide a broader framework for educational technology practice, focusing on the ethical design, use, management, and evaluation of learning resources and environments. AECT highlights the importance of instructional design, resource management, and assessment in integrating technology effectively. Applying AECT principles ensures that the tools introduced in this plan are not only accessible and pedagogically sound but also sustainable and contextually appropriate for a rural school with limited infrastructure (AECT, 2023).

These professional standards ensure that the technology plan is grounded in recognised best practices. They provide a clear framework that supports ethical use, effective teaching, and meaningful learning while addressing the challenges specific to rural schools.

5. Privacy Guidelines

Because the school is new to using digital tools, it is important to create clear and simple privacy guidelines that protect both learners and teachers. These guidelines will help everyone understand how to use Google Classroom, simulations, and virtual learning resources safely and responsibly.

When setting up Google Classroom accounts, only essential information will be needed. Learners will not share personal details such as phone numbers, home addresses, or identification numbers. Teachers will create the class codes and share them directly with learners so that access stays controlled and secure. Learners will also be taught not to share these class codes with anyone outside the school.

All communication on Google Classroom must stay professional and school related. This protects learners from inappropriate conversations and keeps classroom activities focused on learning. Teachers will monitor the platform daily to make sure that comments, submissions, and shared materials follow school expectations.

Since learners may be using shared devices at home, privacy becomes even more important. They will be reminded to log out of Google Classroom after each use, especially when using a public or shared device. They will also be taught not to save passwords on devices that others can access.

For PhET simulations, virtual labs, and virtual museum tours, learners will access these through links provided by their teachers. These tools do not require students to create accounts, which helps keep personal data safe. Teachers will check that all websites used in lessons are secure, appropriate for school use, and free of harmful content.

Lastly, teachers and learners will receive basic guidance on responsible digital behavior. This includes respecting other people's work, not copying materials without permission, giving credit when needed, and reporting any suspicious or uncomfortable online experiences. These simple privacy rules will help the school build a safe digital learning environment, even with limited resources.

6. Artificial Intelligence Policy

This technology plan does not include the direct use of Artificial Intelligence (AI) tools. The main tools being introduced are Google Classroom, PhET simulations, virtual labs, and virtual museum tours. These do not require or depend on AI for teaching and learning activities.

Even though AI is not part of the current plan, it is still important for the school to have a simple policy in place. Technology is changing quickly, and teachers or learners may come across AI tools online, especially when accessing Google services or educational resources. A basic AI policy will help guide responsible and safe behavior if AI tools are encountered or become available in the future.

i. Teachers and learners should not use AI tools to complete assignments, tests, or practical tasks unless a teacher gives specific permission and explains how the tool can be used fairly. This helps prevent plagiarism and ensures that schoolwork reflects the learner's own understanding.

ii. Any AI-generated content, such as text, images, or explanations, must be checked by a teacher before being used in class activities. This is because AI tools can sometimes provide incorrect information, especially in science subjects that require accuracy.

iii. No personal information should be entered into any AI tool. Learners must not share their full names, school details, addresses, photos, or any private information with online AI platforms. This protects their safety and privacy.

iv. If AI is introduced in the future, the school will provide teachers with training first. This training will help teachers understand how AI works, how it can support learning, and how to make sure it is used ethically and responsibly. Only after teachers are confident and prepared will learners be allowed to use AI tools for educational purposes.

7. Equity and Accessibility Considerations

This plan is guided by the ideas in the 2024 National Educational Technology Plan, particularly its emphasis on closing the digital divide. The central priority is ensuring that all learners, regardless of their background, have an equal opportunity to access digital tools, experience interactive science learning, and build the foundational digital skills that will support them beyond high school (U.S. Department of Education, 2024).

Because KwaNxusa High School is a rural, low-resource school, it is important that the technology plan supports fair access for all learners. Many students do not have personal devices, reliable internet, or experience using digital tools. For this reason, the plan focuses on tools and strategies that can work even with limited resources and that ensure no learner is left behind.

To begin with, the main tool, Google Classroom, will be used in a very flexible way. Teachers will upload notes, worksheets, and simple activities that can be downloaded whenever learners have access to a device or network. Assignments will not require constant internet access. Learners can download materials, work offline, and upload when they are back at school or have connectivity. This helps make digital learning possible even for students with limited data.

This helps bridge the digital access divide by making digital learning possible even for students who have limited data.

Simulations, virtual labs, and virtual museum tours will mainly be used during classroom lessons, where the teacher controls the screen using the school's projector or shared device. This ensures that every learner can access the experience together, even if they do not have individual devices. When possible, these tools will be downloaded beforehand so they can run offline without relying on internet during lessons. This also helps reduce the digital use divide, because learners will still engage with digital content even without personal devices.

The plan also considers different learning needs. Visual demonstrations from simulations can support learners who struggle to understand abstract scientific ideas through text alone. Virtual labs and videos give learners with limited laboratory experience the chance to see experiments they would never access in real life. Teachers will provide written notes, voice explanations, and visuals to make sure all learners can understand the content. These decisions are guided by the Universal Design for Learning (UDL) framework, which emphasises offering multiple ways for learners to access information, engage with content, and show what they know (CAST UDL Guidelines, 2025).

To support digital equity, teachers will first receive training on how to introduce technology slowly and clearly. This helps ensure that learners are guided step-by-step and not overwhelmed. The training will include strategies based on the TPACK framework, helping teachers learn how to integrate technology in a way that fits their teaching methods and the Science content they teach (Mishra & Koehler, 2006).

8. Required Personnel

To successfully execute this plan, the school will need a small but committed team of people who will support the rollout.

First, the main group needed is the Science Department teachers. These are the teachers who will use the technology tools directly in their classes. Their role will be to create and manage tools during lessons.

Second, the school will need one Technology Lead Teacher. This does not have to be a specialist or someone with formal IT training. It can simply be a teacher who is comfortable with digital tools and willing to learn. This person will help other teachers set up Google Classroom accounts and troubleshoot basic issues.

Third, the school principal plays an important role. The principal will provide overall leadership, approve the plan, and ensure that teachers have the time, resources, and support they need (easing the digital use divide). The principal will also help communicate the purpose of the technology plan to parents, school management teams, and district officials.

Fourth, the school will need occasional support from the district IT or e-learning office. This may include helping to set up Wi-Fi, advising on device maintenance, or offering teacher workshops when possible.

Finally, learners themselves are an important part of the personnel needed. They will learn how to use the tools responsibly, support one another, and give feedback to teachers about what works well and what needs improvement. In many schools, learners quickly become confident with technology and can help peers who are struggling.

9. Budget

KwaNxusa High School is a no-fee, rural school with limited funding, so this technology plan focuses on using tools that are free, low-cost, and easy to maintain. The goal is to introduce essential digital learning tools without placing financial strain on the school.

Table 1: 1 year plan implementation budget

Category	Item/Service	Unit Cost (US$)	Quantity	Total Cost (US$)	Notes
Hardware	Ultra-Link Smart Compact Projector	$290.61	8	$2324.88	One projector per science classroom
	Samsung 27" Full HD Monitor 100Hz	$177.18	8	$1417.44	One monitor per science classroom for teacher podium
	JBL GO Essential 2 Portable BT Speaker	$41.30	8	$330.40	One speaker per science classroom

Software & Tools	Google Workspace for Education	Free	–	$0	Free version suitable for low-resource schools
	PhET Interactive Simulations	Free	–	$0	Downloadable for offline use
	Virtual Labs & Virtual Museum Tours	Free	–	$0	Open-access resources
Connectivity	Mobile data bundle for school device	$15 per month	12 months	$180	Used to download materials and updates; offline use thereafter
Training & Support	Teacher training workshop	$295.40	1	$295.40	ICT orientation and pedagogical support
Maintenance	Device repair/contingency	$295.40	1	$295.40	Estimated repairs and maintenance for the year
TOTAL ESTIMATED COST				**$4,876.20**	Year 1 implementation

Budget Summary Explanation

This budget reflects the essential costs required to introduce basic technology infrastructure into the Science department at KwaNxusa High School. Each classroom receives one projector, monitor, and speaker to support interactive teaching using Google Classroom, PhET simulations, and virtual learning resources. All selected tools are free to use, which helps keep the overall cost low for a no-fee rural school. A mobile data bundle is included for

downloading materials and updating the system, while most teaching will occur offline. The plan also includes one teacher training workshop to ensure effective use of the tools, and a small maintenance allocation for repairs. Overall, the budget represents a realistic, low-cost approach for a resource-constrained rural school implementing technology for the first time.

10.Potential Obstacles

There are several challenges that the school may face during implementation. Understanding these obstacles early helps the school plan realistic solutions.

i. Limited Access to Internet
ii. Shortage of Digital Devices
iii. Teachers' Digital Skills and Confidence
iv. Technical Issues and Lack of IT Support
v. Learners' Digital Literacy
vi. **Sustainability Over Time** - Maintaining Wi-Fi payments, repairing devices, and keeping tools updated may be challenging if the school lacks ongoing funding. Without long-term planning, the technology integration may slow down or stop after the first year.
vii. **Possible Resistance to Change** - Some teachers, learners, or even parents may prefer traditional textbook-based learning. They may not immediately see the value of digital tools. Overcoming this requires patience, training, and clear communication about the benefits.

11.Timeline for Implementation

The following table outlines the step-by-step process for introducing and fully implementing the selected technology tools at KwaNxusa High School over a 12-month period. The plan begins with preparation and purchasing, followed by teacher training, a short pilot phase, full-scale implementation, and finally a review stage to assess progress and identify improvements. The goal is to introduce technology gradually and sustainably, ensuring that teachers feel confident and learners benefit from consistent access to digital Science resources.

Table 2: Timeline for Implementation

Phase	Timeframe	Key Activities	Responsible Personnel
1. Preparation & Planning	Month 1	• Finalise technology plan • Confirm budget and procurement list • Meet with Science teachers to review roles • Prepare school device for data and downloads	Principal, ICT Lead Teacher, Science HOD
2. Procurement & Setup	Month 2	• Purchase projectors, monitors, speakers • Install devices in all 8 science classrooms • Download PhET simulations and virtual lab resources for offline use • Set up Google Workspace accounts for teachers	ICT Lead Teacher, Technician, Administration
3. Teacher Training	Month 3	• Conduct one full-day training workshop • Cover Google Classroom setup, offline PhET use, virtual resources, digital safety • Provide training materials and support templates	External Trainer / ICT Lead Teacher
4. Pilot Implementation	Months 4–5	• Teachers introduce Google Classroom with basic tasks	Science Teachers, ICT Lead Teacher

		• Use simulations and virtual labs during lessons • Gather feedback from teachers and students • Identify challenges and required adjustments	
5. Full Implementation	Months 6–9	• Regular use of Google Classroom for assignments and resources • Weekly use of projectors and simulations in lessons • Provide ongoing teacher support and troubleshooting • Begin monitoring student engagement and progress	Science Teachers, HOD, ICT Lead Teacher
6. Review & Improvement	Months 10–12	• Review implementation successes and challenges • Update teaching practices based on feedback • Evaluate the use of digital tools and learner progress • Plan next-year improvements	Principal, Science HOD, ICT Lead Teacher

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